

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Friday, 01/31/2020

Quiz 1: Section 2.1, 2.2

Name (Print): _____

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Section 2.1

2 pts

1. Suppose you throw a rock straight upwards with a speed of $96 \frac{\text{ft}}{\text{s}}$. Neglecting air resistance, a well-known formula from physics states that the position of the rock after t seconds is given by the function

$$s(t) = -16t^2 + 96t.$$

What was the *average velocity* of the rock between $t = 1$ s and $t = 4$ s?

Using the definition of average velocity, the average velocity of the rock between $t = 1$ s and $t = 4$ s is

$$\begin{aligned} v_{\text{avg}} &= \frac{(s(4) - s(1)) \text{ ft}}{(4 - 1) \text{ s}} = \frac{(-16(4)^2 + 96(4)) - (-16(1)^2 + 96(1)) \text{ ft}}{3 \text{ s}} \\ &= \frac{128 \text{ ft} - 80 \text{ ft}}{3 \text{ s}} \\ &= \frac{48 \text{ ft}}{3 \text{ s}} = 16 \text{ ft/s.} \end{aligned}$$

2 pts

2. Let $f(x) = x^2 - 5$. Find the slope of the secant line between $(1, f(1))$ and $(5, f(5))$.

Using the definition of a secant line, the slope of the secant line passing through the points $(1, f(1))$ and $(5, f(5))$ is

$$\begin{aligned} m_{\text{sec}} &= \frac{f(5) - f(1)}{5 - 1} \\ &= \frac{(5^2 - 5) - (1^2 - 5)}{4} \\ &= \frac{20 - (-4)}{4} = \frac{24}{4} = 6. \end{aligned}$$

Section 2.2

3 pts

1. If $f(x) = \begin{cases} x+2 & \text{if } x \leq 0 \\ 2-x & \text{if } x > 0 \end{cases}$, find $\lim_{x \rightarrow 0} f(x)$.

For the limit to exist, it must match up with the left-hand and right-hand limits. First, we calculate the left-hand limit:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x + 2 = 0 + 2 = 2.$$

Then, we calculate the right-hand limit:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 2 - x = 2 - 0 = 2.$$

As these are equal, we then yield that

$$\lim_{x \rightarrow 0} f(x) = 2.$$

3 pts

2. Find $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$ or explain why the limit does not exist.

If we try to calculate the limit directly, we yield

$$\lim_{x \rightarrow 3} \frac{x-3}{x^2-9} = \frac{3-3}{(3)^2-9} = \frac{0}{0},$$

which is an indeterminate form. So, we must factor and cancel to yield the correct solution:

$$\lim_{x \rightarrow 3} \frac{x-3}{x^2-9} = \lim_{x \rightarrow 3} \frac{x-3}{(x+3)(x-3)} = \lim_{x \rightarrow 3} \frac{1}{x+3} = \frac{1}{3+3} = \frac{1}{6}.$$